

Service Classification in Ericsson UPF



Walk through Ericsson UPF architecture capabilities on Traffic / Service classification and discuss the Demo logistics

Agenda

- Best Network – Big Ideas
- Service Recognition Use Case
- Service Classification capabilities in the UPF
- Demo Scenarios and Logistics

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Most AI-aware network – Ideas

Best “AI
ready”
network



With UE OEM

- Delay Status Reporting
(IODT 271H)
 - UE sends delay reports for latency sensitive applications
 - Benefits AI applications like Siri, or Smart glasses on best effort bearer yet is latency sensitive to improve user experience
- L4S Enhancements
(In planning)
 - RAN sends L4S congestion info directly to the UE and avoids round trip delay
 - Improves user experience for latency sensitive applications (AI glasses, mobile gaming, etc).
- ..and more

Device agnostic E2E RAN-Core Intelligence

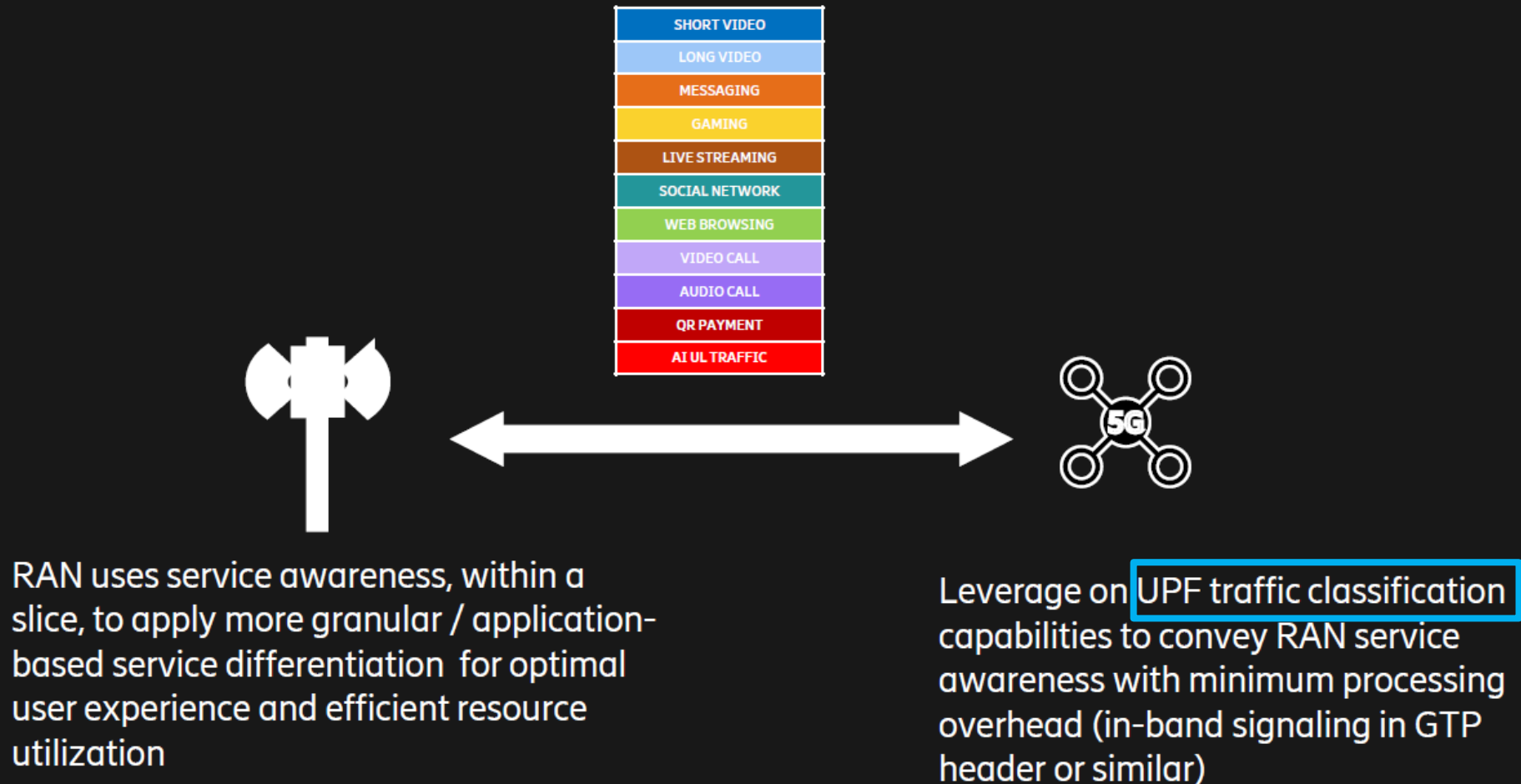
- Service Recognition
(Concept exploration with AT&T)
 - Identify in real-time, application traffic within a slice and offer differentiated handling
 - Mitigates dependency on UE OS w.r.t. URSP for traffic differentiation
 - Benefits use-cases like FirstNet – where in real-time, any “non-prioritized” traffic (e.g. streaming video) can be identified from “prioritized mission critical traffic” and handled according to ATT policy (e.g. down prioritize)
 - Other use-cases incl. RoboTaxi , OTT Audio/Video calling, etc
- ...and more
 - Mobility Prediction
 - Traffic Prediction

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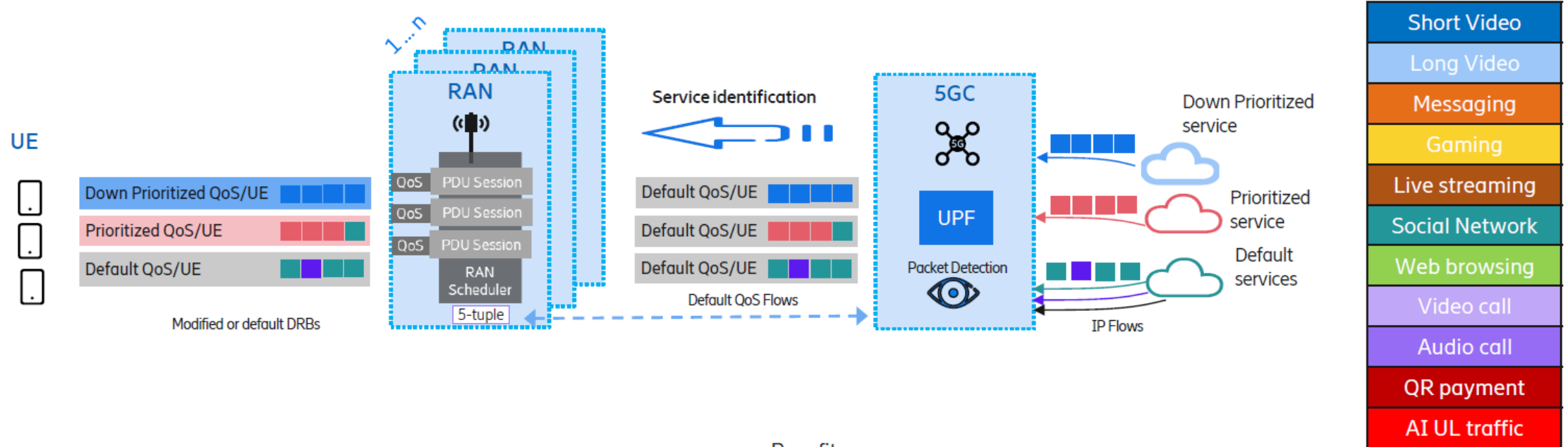
Service Awareness driven User Experience

Complement to E2E Slicing deployment



Service/Application/Flow recognition for RAN Efficiency

UPF providing service classification to RAN for Service Awareness handling in RAN



● Solution

- Leverage PCG traffic inspection to identify different service type
- Ericsson unique inline signaling between Core and RAN to convey service type from Core to RAN
- RAN uses service type, within a slice, to apply application-based service differentiation
- Support for both heuristics and deterministic services

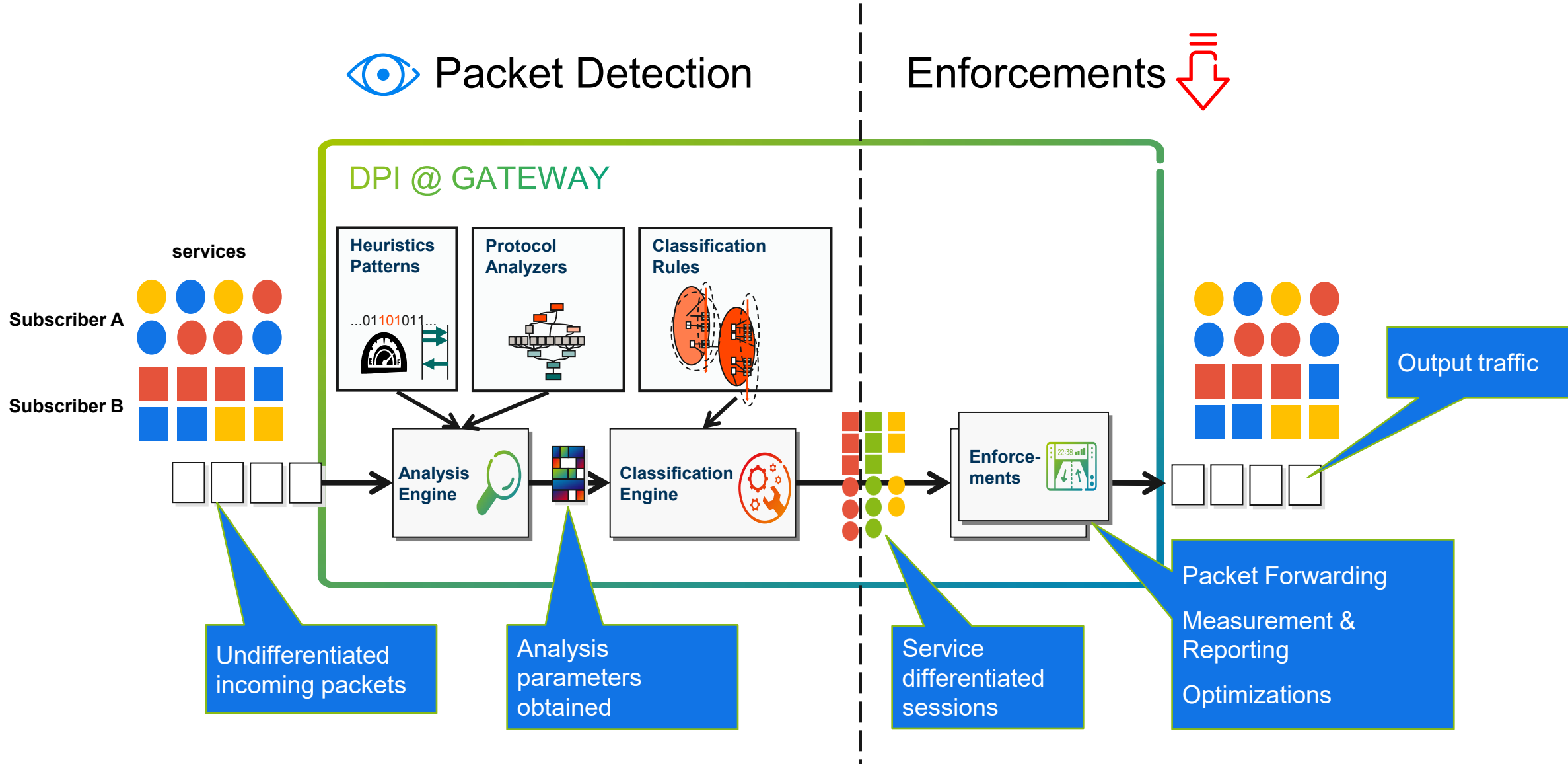
● Benefits

- Minimize processing overhead by reusing traffic recognition in PCG
- Operator decision about what type of traffic and for which subscribers to prioritize or down prioritize
- High traffic accuracy 95% traffic for 1000+ apps
- Online Upgrades of traffic classifications
- Maintains service classification also during Inter gNB mobility

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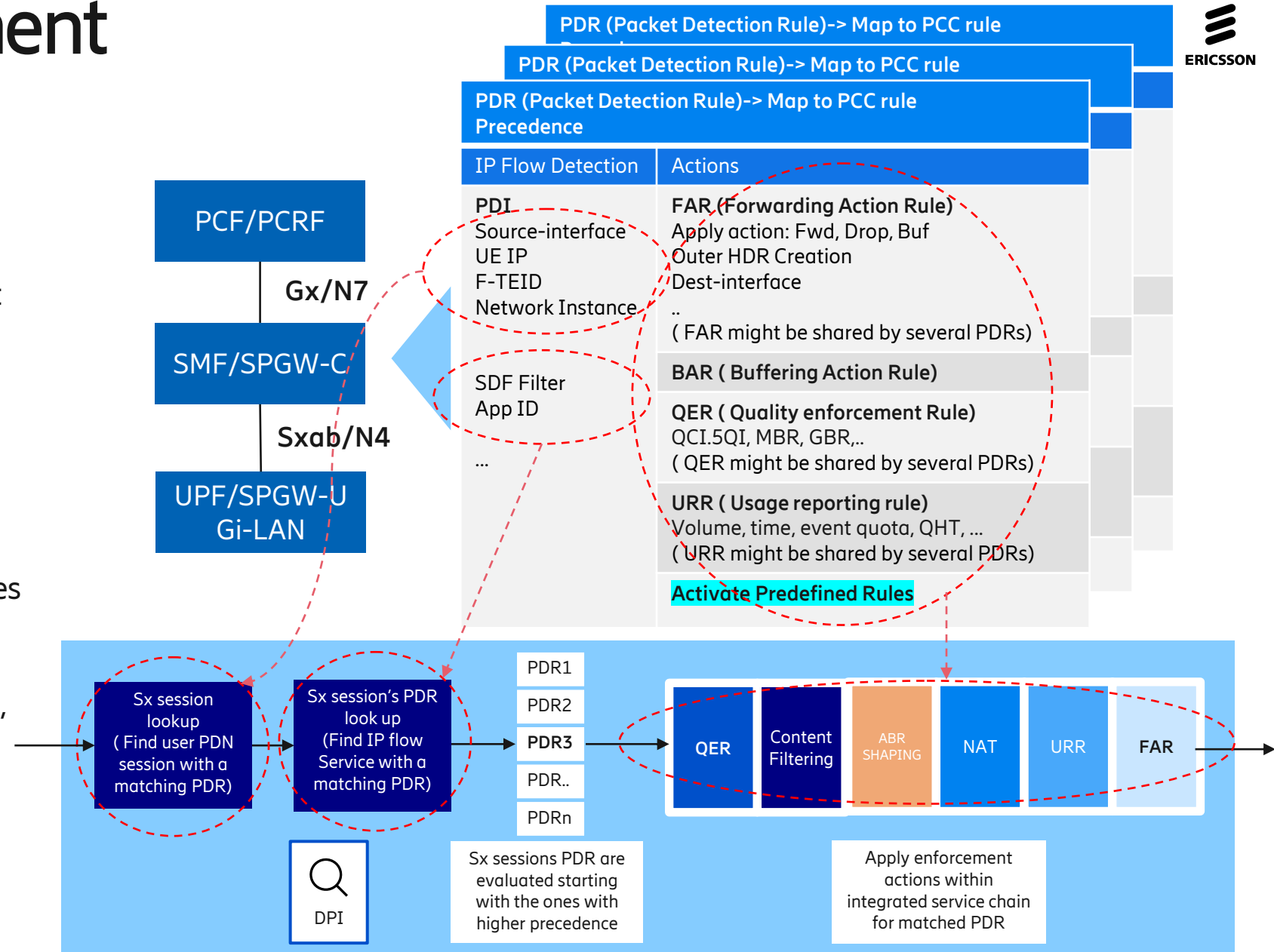
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Packet Classification and Enforcements

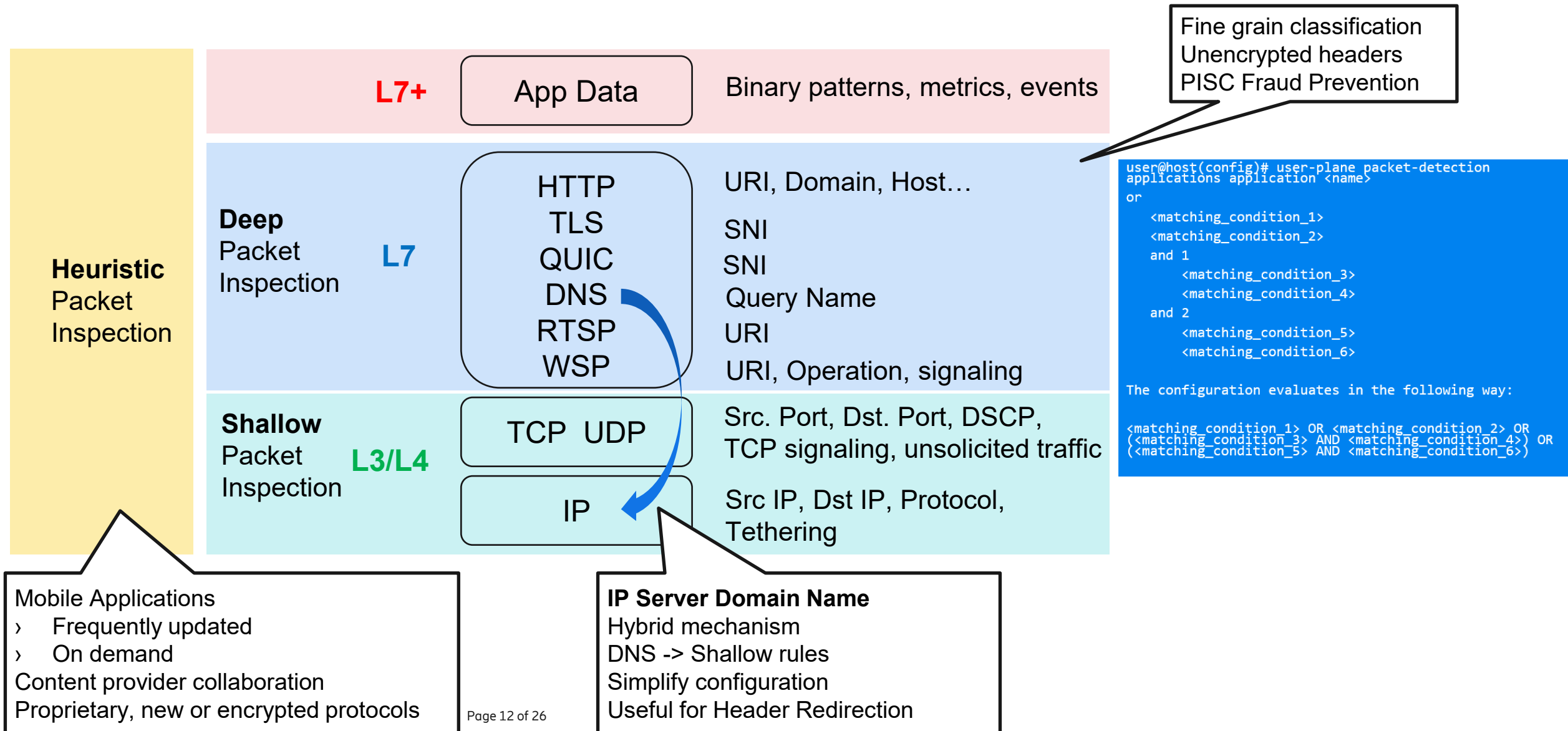


Traffic management

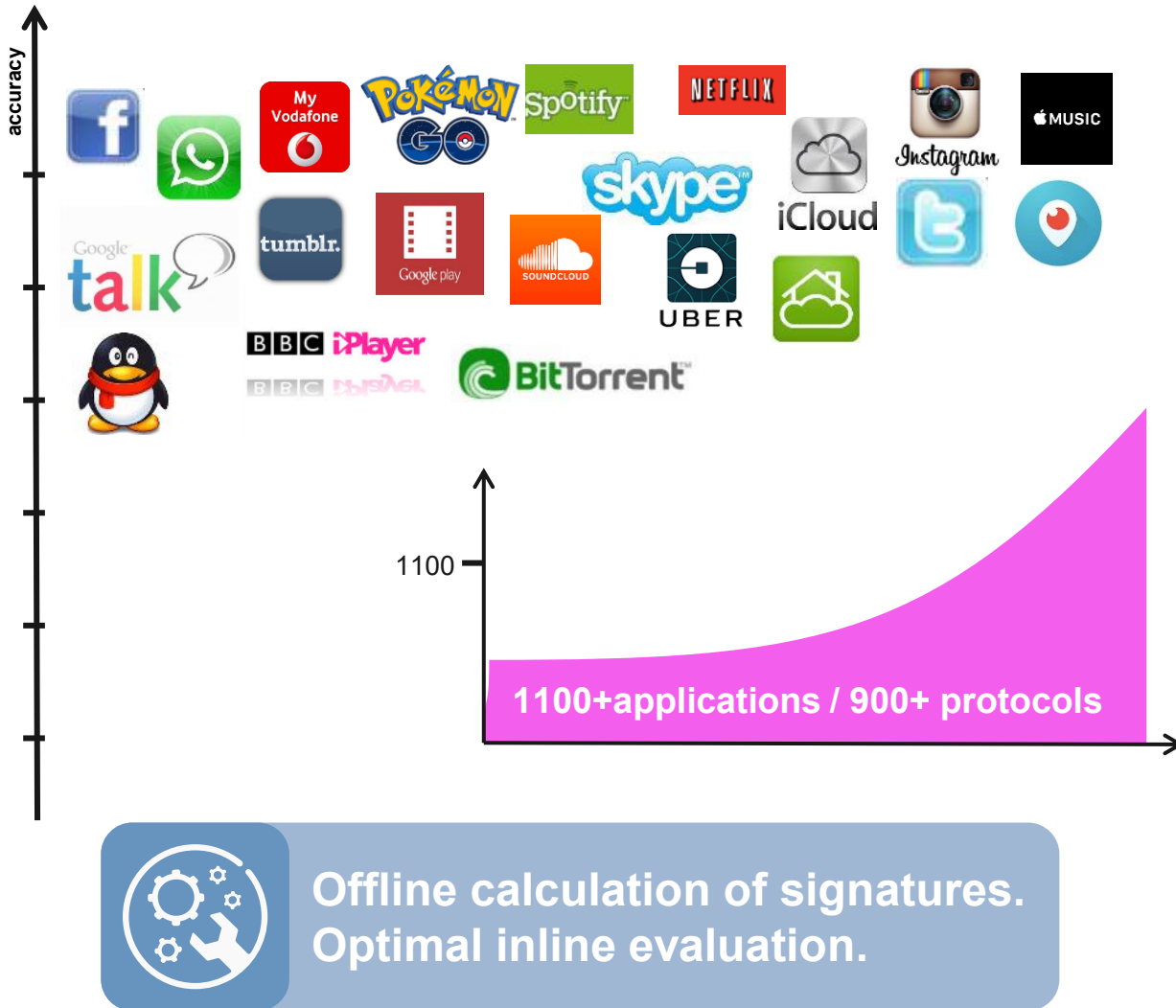
- Filters and actions might be provision dynamically by Control Plane or might be preconfigured in User Plane (memory and signaling efficiency)
- Non 3GPP actions (Video/traffic optimization, Content filtering, NAT/FW ...) can be invoked through standard N7/Gx, Sx/N4
- PCG Integrated Service chain combines flexible 3GPP actions (FAR, BAR, QER, URR) and non 3GPP actions (i.e traffic / video optimization, content filtering, NAT , Firewall)



Packet Analysis & Classification Capabilities



Heuristics Classification



Flexible Classification



Ready to Deploy



High accuracy



95%+ of accuracy

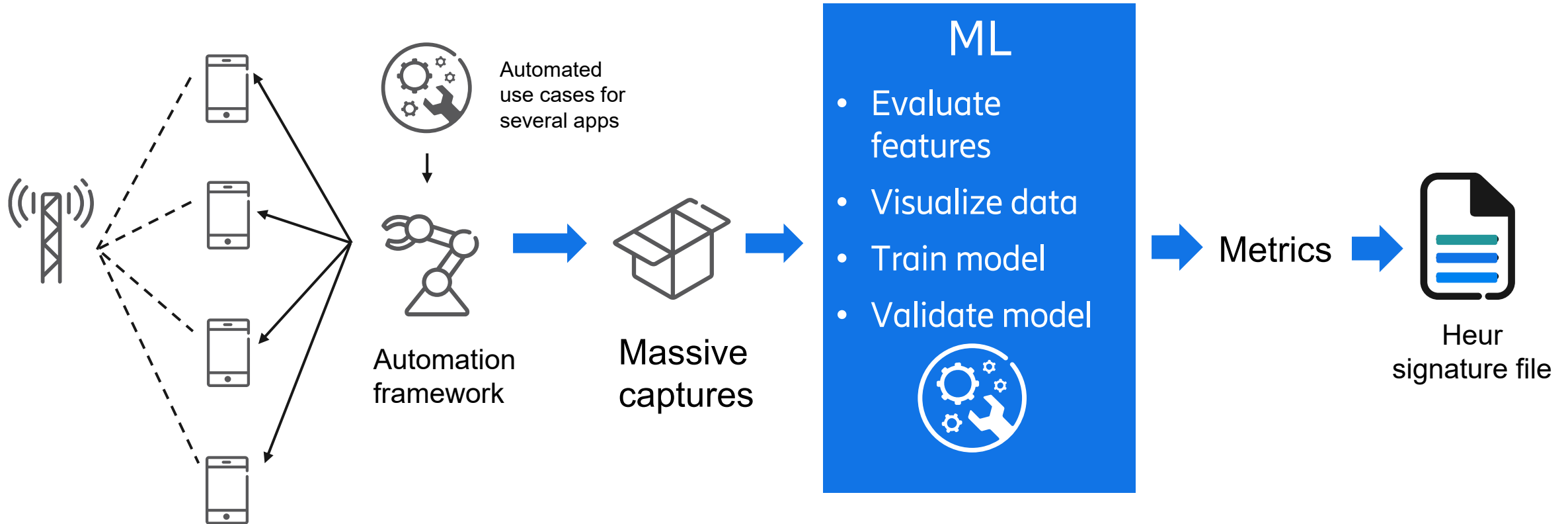


1wk release cycle



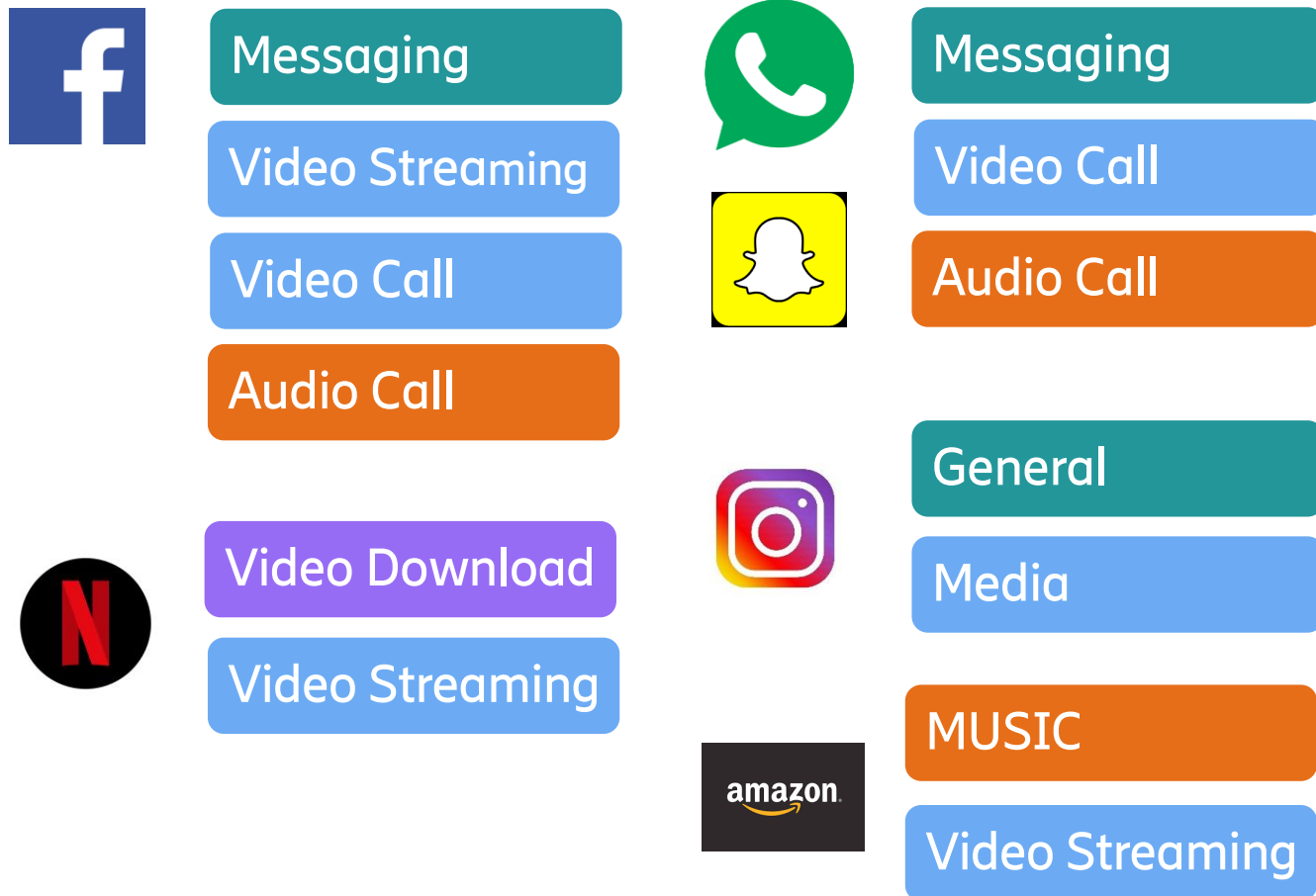
Hot updates

Machine Learning Workflow



Differentiation within APPs reinforced with ML based models

To enable commercial offerings targeting specific types of user activities a number of popular signatures have the option of differentiation by traffic type.



- Ericsson applies ML in different areas. Service Classification is one of the priority areas addressed, where we already have new signature using ML added in our traffic detection engine in EPG/PCG.
- ML techniques are mostly used to differentiate flow streams within applications where existing HPI methods are not valid.
- We have multiple ML developed algorithms for traffic classification, some of them already included in the Ericsson products classification portfolio (inline classification) and could be reused by Analytics.
- We are also exploring other techniques like neural networks applied to traffic classification (when no SNI) and showed to be a track worth exploring.



AI

Artificial Intelligence Detection



Protocols included in the AI Group:

- OpenAI
- Claude
- Copilot
- Gemini
- Deepseek
- Grok-ai
- Perplexity
- Midjourney
- Galaxy-AI

Domain-based classification across QUIC and TLS



The Heuristics Engine recovers the QUIC Initial ClientHello in real time — and uses the same domain signal exposed by standard TLS.



Scope: QUIC Initial only; subsequent QUIC payload remains protected. ECH can hide the true SNI.

Sources: RFC 9001 §5.2–5.4, 8.1 · RFC 9000 · RFC 6066 §3 · RFC 9849 (ECH)

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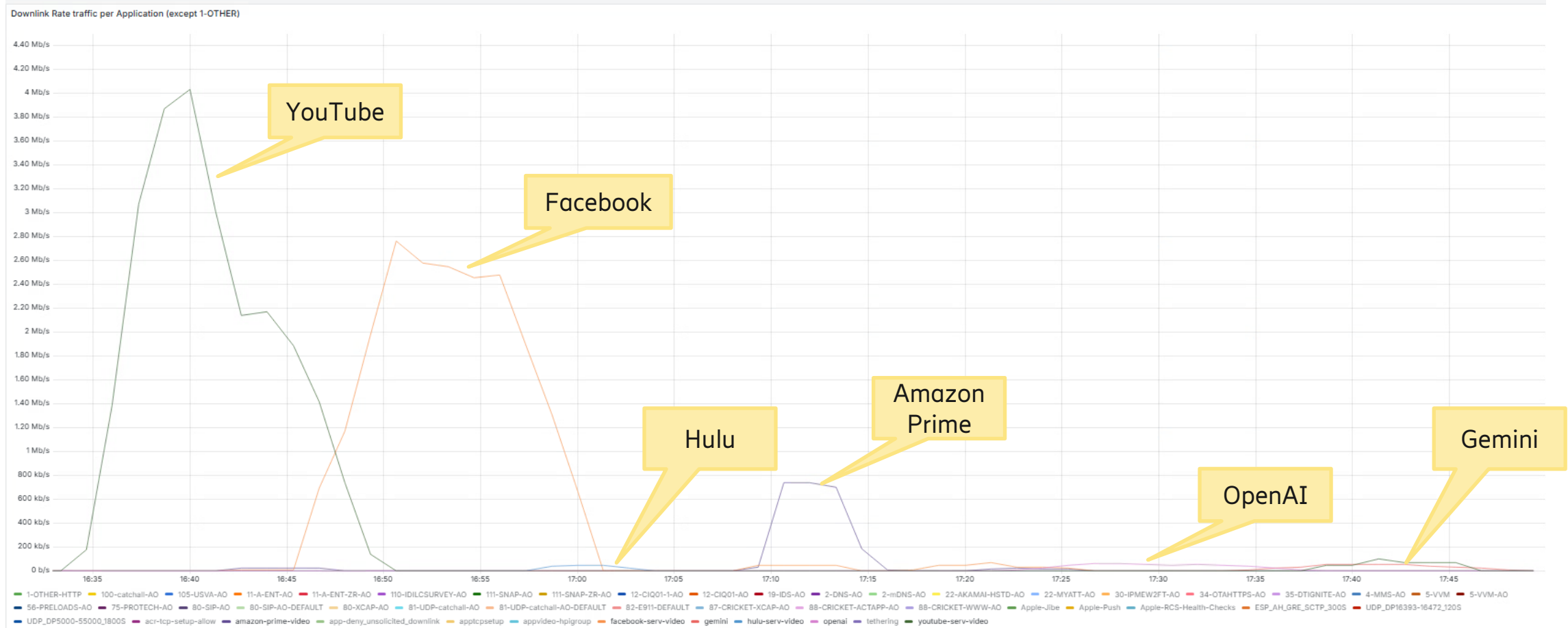
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Demo

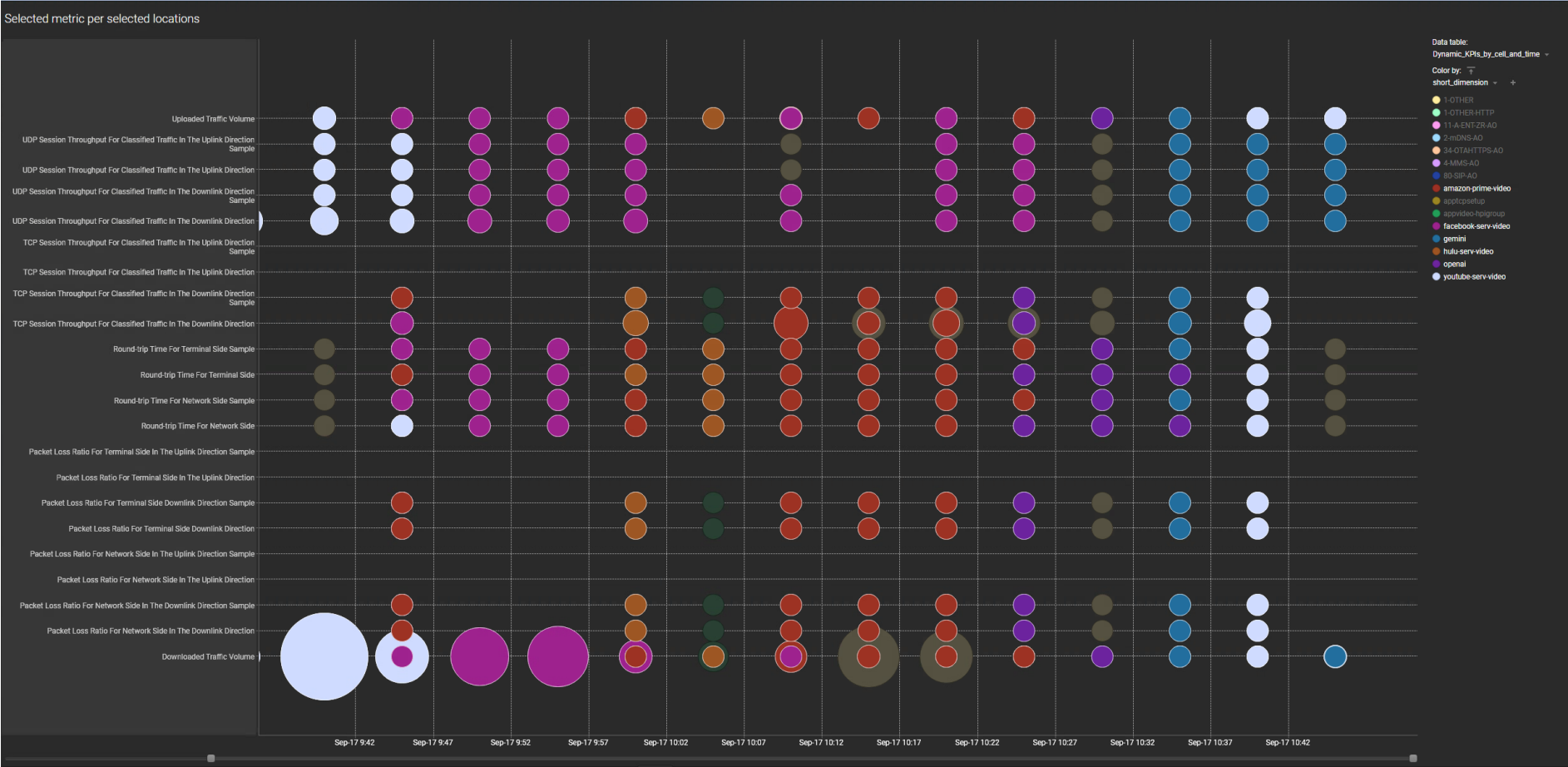
- Demo Scope
 - Show the classification of services (YouTube, Facebook, Hulu, Amazon Prime, OpenAI, Gemini,) from the Test UE Traffic
- Demo Lab
 - AT&T Redmond CTO Lab

- Demo Scenarios
 1. Demo lab setup
 2. How the traffic is simulated with Test UE
 3. Show the service classification

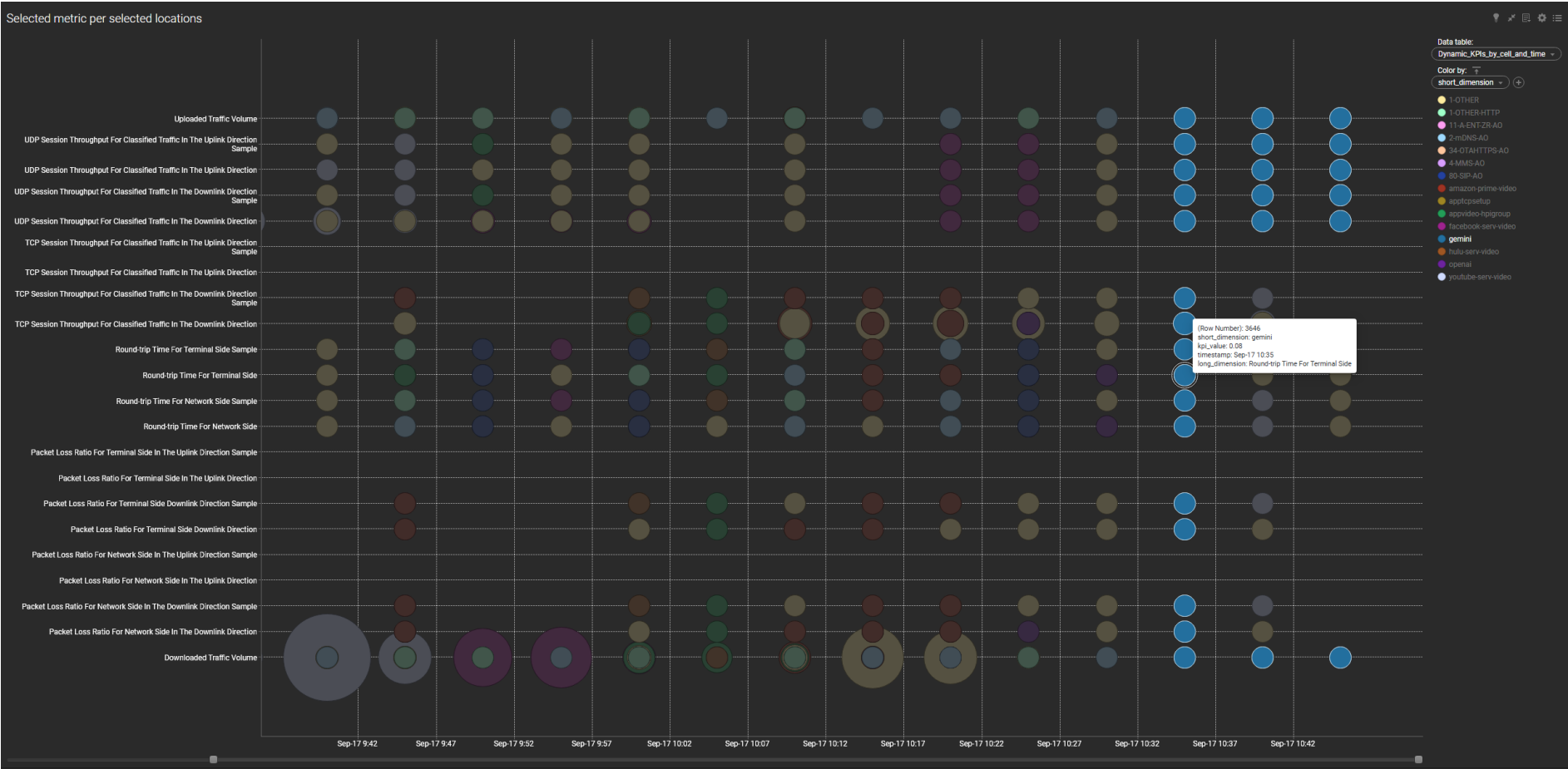
PCG Observability – Service Classification



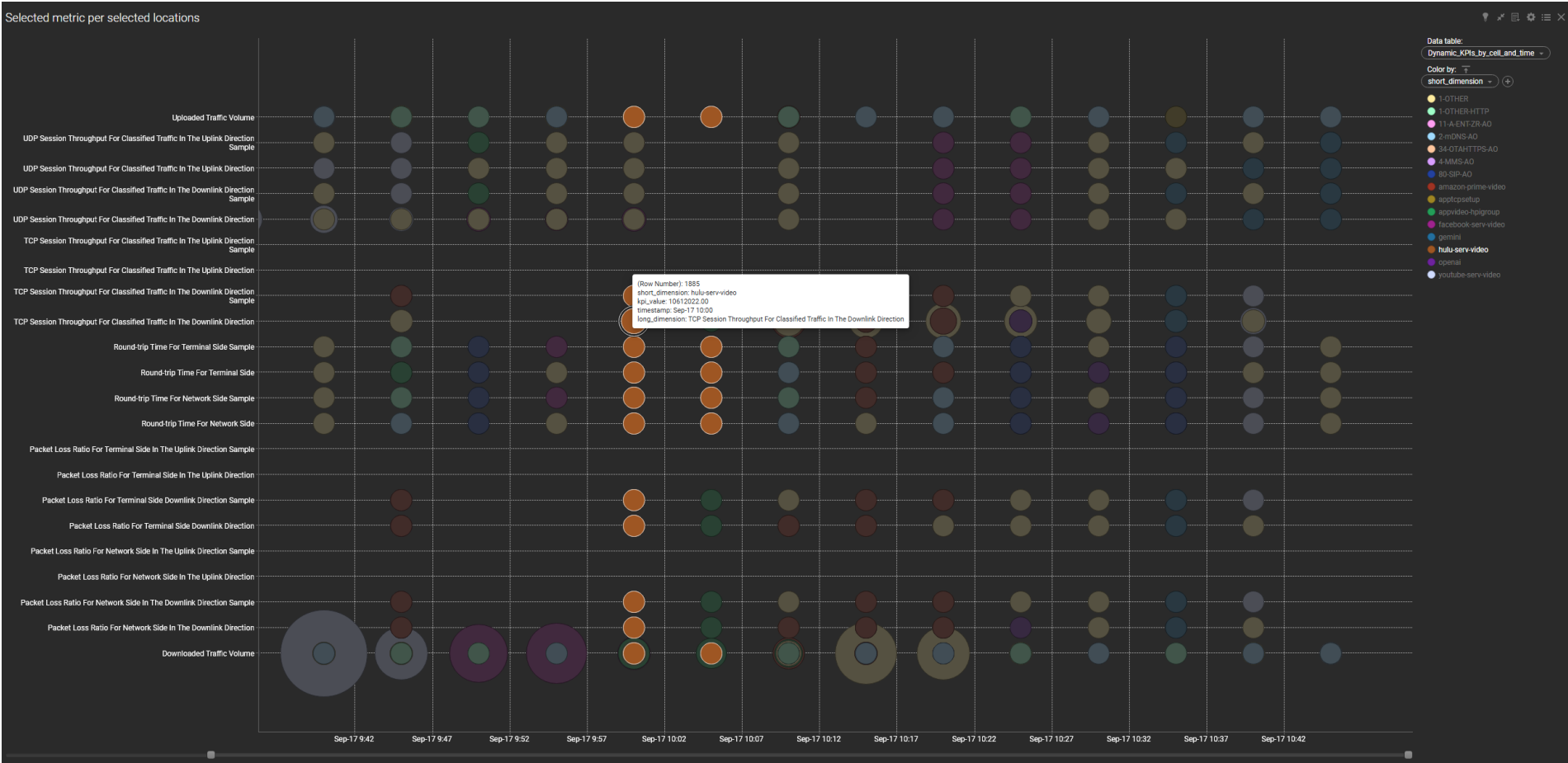
EEA Observability – All Applications - Service Classification



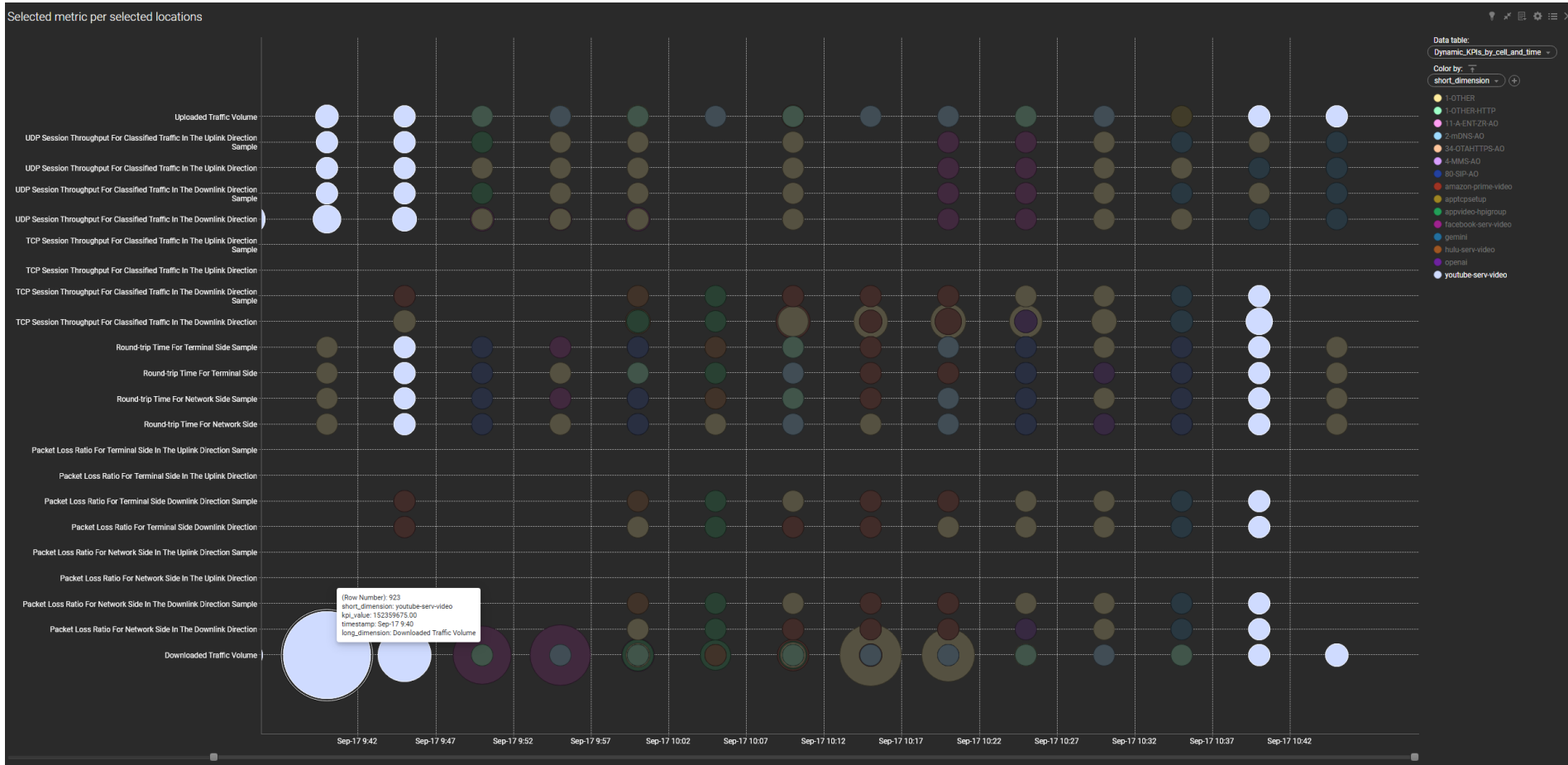
EEA Observability – Gemini Service Classification



EEA Observability – Hulu Service Classification



EEA Observability – YouTube Service Classification





150 years